DATA SCIENCE CAPSTONE PROJECT

SpaceX Falcon 9
First-Stage Landing Prediction

Exploratory analysis • SQL • Folium • Plotly Dash • Classification

Project repository

https://github.com/MatiasMontenegrom/spacex-falcon9-landing-prediction

Evidence-based revision | Screenshots and notebook outputs supplied for the project

Executive summary

What this project delivers and what the evidence supports

- Combines launch-record exploration, SQL aggregation, geographic mapping, an interactive dashboard, and supervised classification.
- The dashboard filters launch site and payload-mass range, then compares payload mass with launch outcome and booster version.
- SQL identifies four launch-site labels in the source and returns 619,967 kg as the sum of recorded payload masses.
- Historical model outputs show Decision Tree test accuracy of 94.4% (17 of 18 test examples); this is a small test set, so treat it as preliminary.

4

Distinct site labels

619,967 kg

Total recorded payload mass

94.4%

Decision Tree test accuracy*

18

Test observations*

*Cached notebook result; re-run for final reproducibility.

Introduction & objective

Why predict Falcon 9 first-stage landing success?

- Reusable first stages can reduce launch costs when recovery is successful.
- The project asks whether recorded launch attributes help predict the binary landing outcome.
- Target used in the course dataset: Class (0/1). Predictors include payload mass, launch site, orbit, flight number, and related features.
- Prediction is an analytical aid—not a guarantee of a future landing.

01 **Collect records**

02 **Clean & explore**

03 **Map & query**

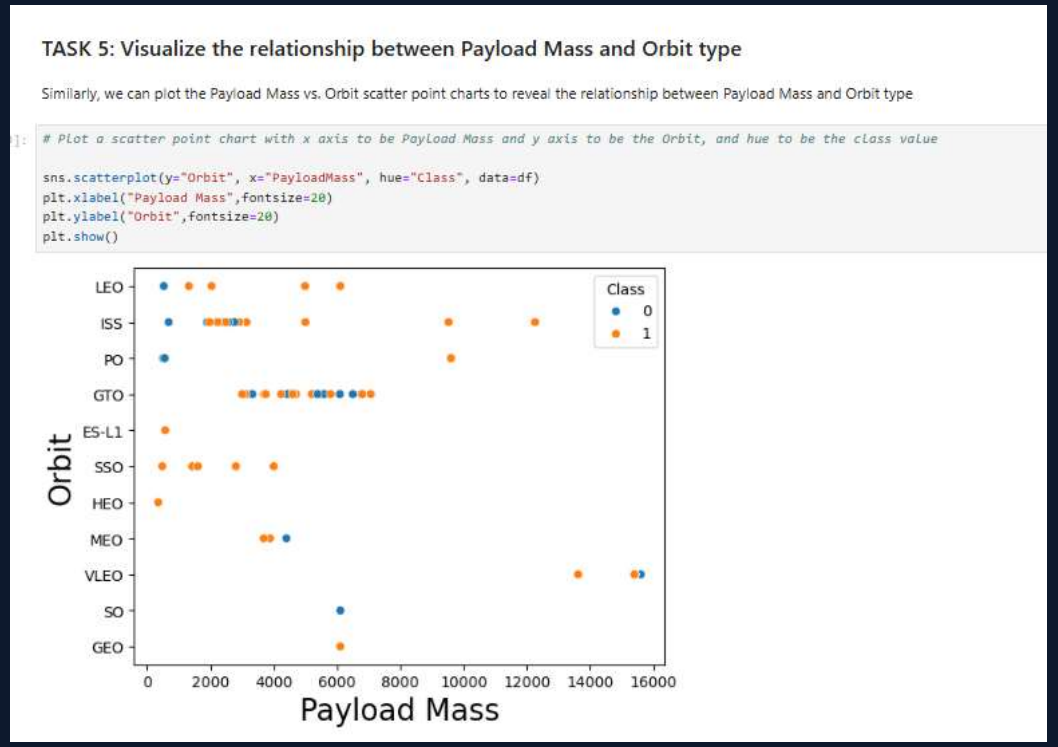
04 **Train classifiers**

05 **Evaluate**

Data collection & wrangling

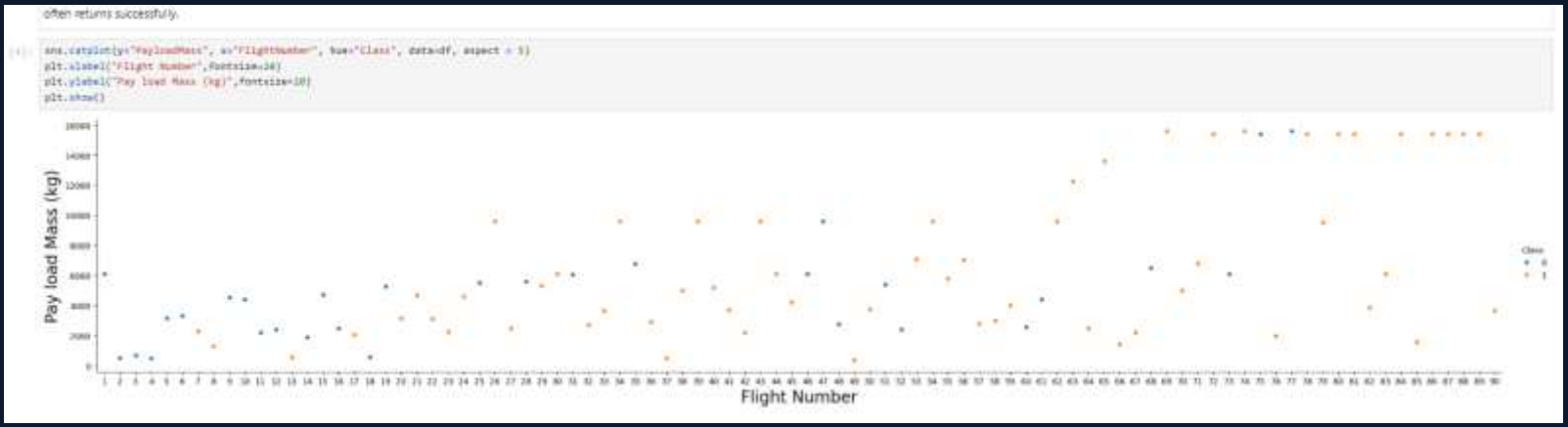
Data fields, transformations, and quality considerations

- Launch records include date, booster version, launch site, payload mass, orbit, mission outcome, and landing outcome.
- The predictive dataset contains 90 numbered flights in the supplied visualizations.
- Class is the target; predictors are kept separate from the target to avoid a KeyError and accidental target leakage.
- Categorical features are encoded; numerical scaling is fitted on training data only.
- Site labels vary (for example, CCAFS LC-40 and CCAFS SLC-40), so standardization is important before aggregation.



EDA overview: payload mass by flight

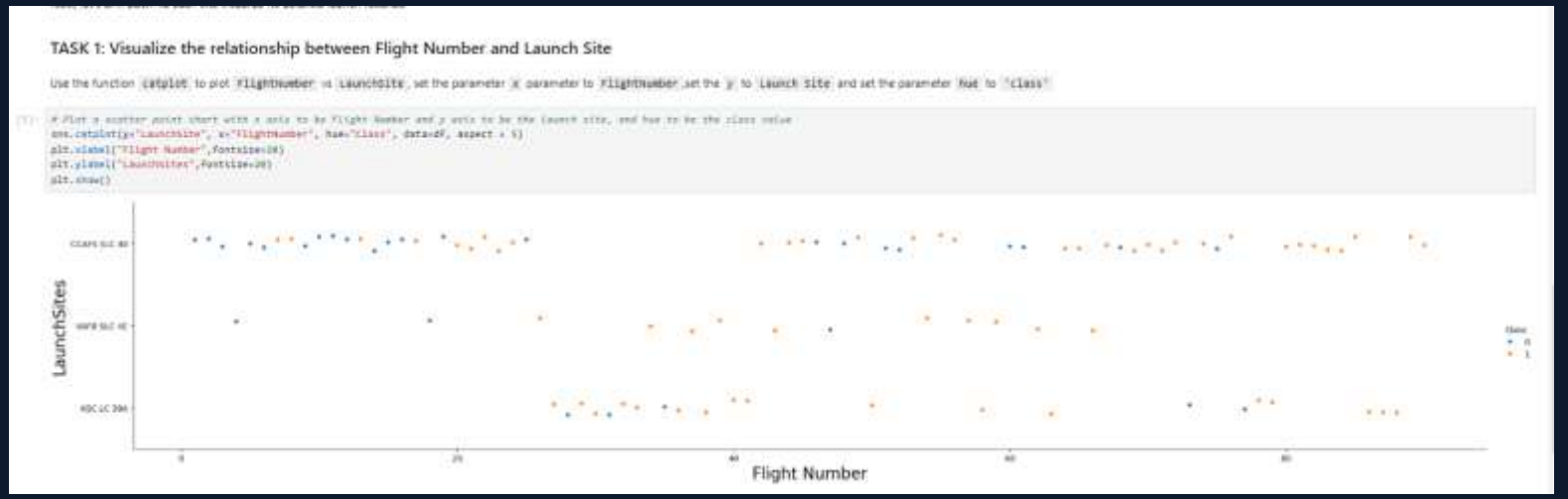
The supplied plot shows a wide range of payload masses and mixed outcomes



- Payload mass varies from very small values to above 15,000 kg.
- Both classes occur across the flight sequence; payload mass alone does not cleanly separate outcomes.
- High-mass missions are present among both successful and unsuccessful outcomes.

EDA: launch site and flight number

Success and failure labels are distributed across multiple sites



- CCAFS SLC-40 has the densest concentration of records in this plot.
- Later flight numbers include more observations at KSC LC-39A and VAFB SLC-4E.
- Because sites have different sample sizes and mission mixes, raw counts should not be interpreted as site-specific causal effects.

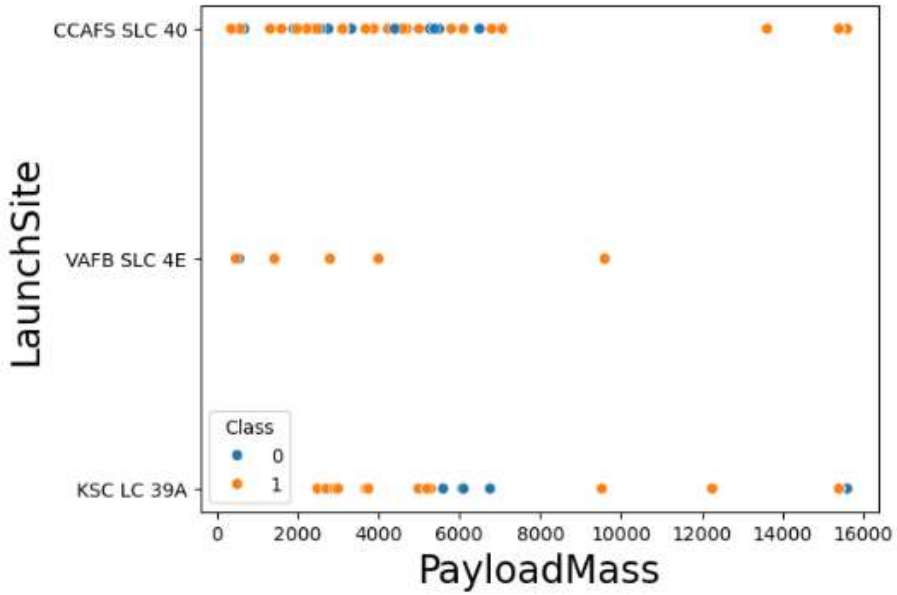
EDA: payload mass by launch site

Payload distributions differ by site in the supplied sample

TASK 2: Visualize the relationship between Payload Mass and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

```
[6]: # Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.scatterplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df)
plt.xlabel("PayloadMass",fontsize=20)
plt.ylabel("LaunchSite",fontsize=20)
plt.show()
```



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than

- The VAFB SLC-4E observations shown do not include payloads above 10,000 kg.
- CCAFS SLC-40 and KSC LC-39A include a broader spread of payload masses.
- Compare rates only after accounting for the number and type of missions at each site.

EDA: observed success rate by orbit

Some orbit categories reach 100% in this sample, but sample sizes matter

Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

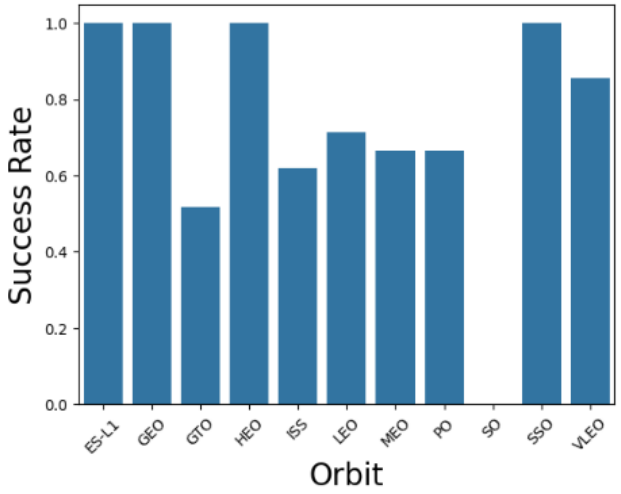
TASK 3: Visualize the relationship between success rate of each orbit type

Next, we want to visually check if there are any relationship between success rate and orbit type.

Let's create a `bar chart` for the sucess rate of each orbit

```
orbit_success = df.groupby('Orbit')['Class'].mean().reset_index()

sns.barplot(data=orbit_success, x='Orbit', y='Class')
plt.xlabel("Orbit", fontsize=20)
plt.ylabel("Success Rate", fontsize=20)
plt.xticks(rotation=45)
plt.show()
```



Analyze the plotted bar chart to identify which orbits have the highest success rates.

- ES-L1, GEO, HEO, and SSO appear at 100% in the chart.
- GTO is near 52%; several other categories fall between roughly 60% and 86%.
- SO has no visible bar, likely because no usable observations appear for that category.
- These are sample proportions, not stable population estimates.

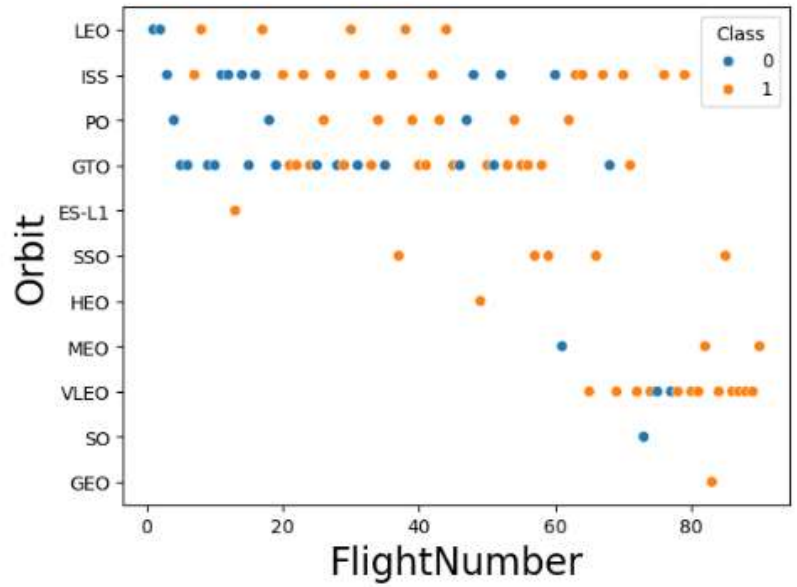
EDA: flight number and orbit type

Flight number appears to co-vary with the types of missions launched

TASK 4: Visualize the relationship between FlightNumber and Orbit type

For each orbit, we want to see if there is any relationship between FlightNumber and Orbit type.

```
[8]: # Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.scatterplot(y="Orbit", x="FlightNumber", hue="Class", data=df)
plt.xlabel("FlightNumber",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```

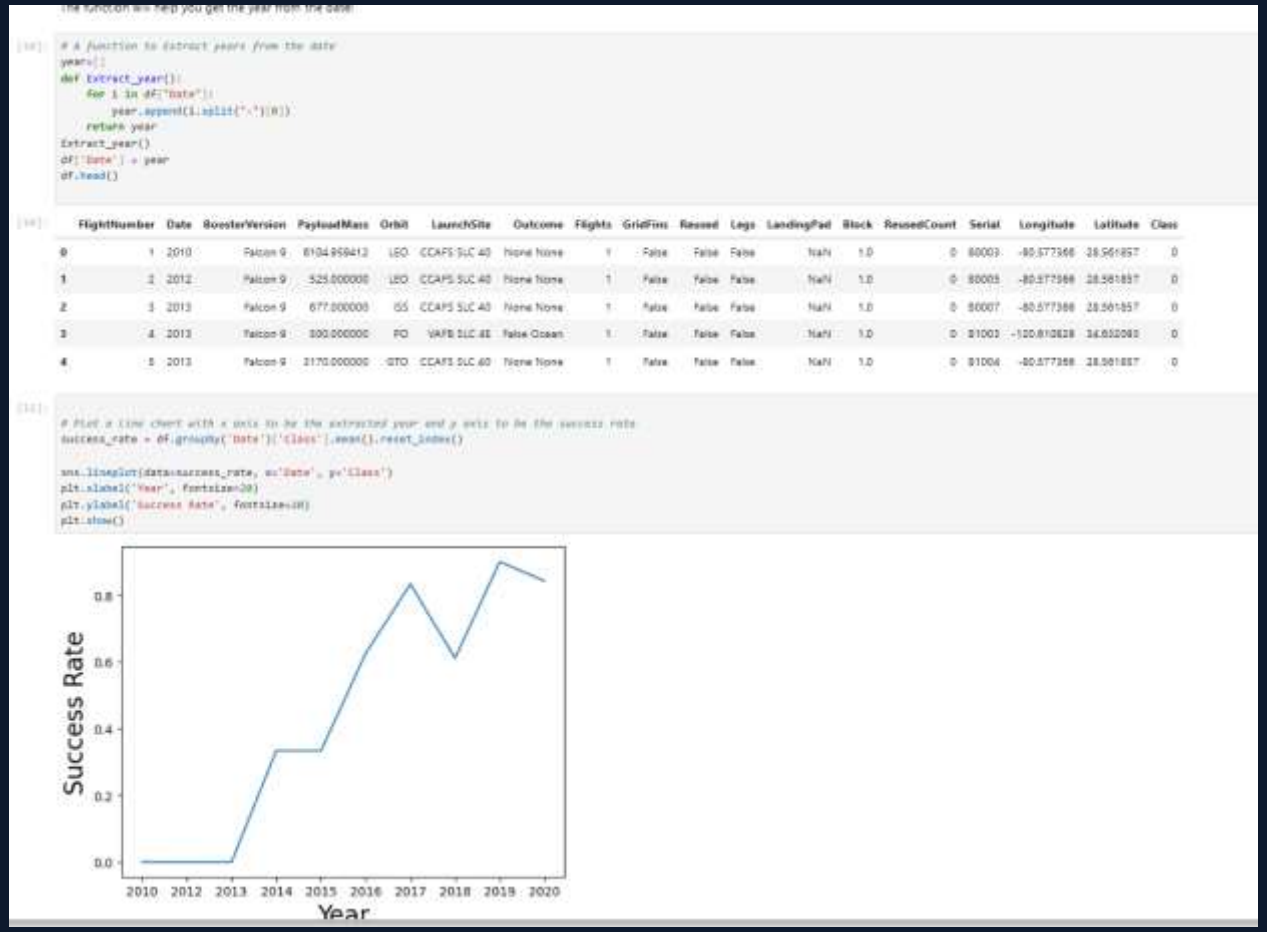


You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight numbe

- LEO and GTO observations span much of the flight sequence.
- VLEO appears concentrated in later flight numbers in this dataset.
- Mission mix changes over time, so a model may learn temporal or mission-type patterns as well as landing-related signals.

EDA: annual success-rate trend

The plotted annual rate rises after the early years



- The supplied chart shows near-zero rates in the earliest years, then a rise from 2014 onward.
- The line reaches approximately 0.83 in 2017, dips to about 0.61 in 2018, and peaks near 0.90 in 2019.
- Early years have very few observations; annual averages can swing sharply when the denominator is small.
- Treat the trend as descriptive, not proof that year itself causes success.

EDA with SQL: query-backed findings

Selected results from the executed SQLite notebook

SQL question	Observed result
Distinct launch-site labels	CCAFS LC-40; VAFB SLC-4E; KSC LC-39A; CCAFS SLC-40
Total payload mass	619,967 kg
First successful ground-pad landing	2015-12-22
Successful drone-ship landings with payload 4,000–6,000 kg	4 booster records shown
Landing outcomes (2010-06-04 to 2017-03-20)	No attempt: 10; Success (drone ship): 5; Failure (drone ship): 5; Success (ground pad): 3; Controlled ocean: 3; Uncontrolled ocean: 2; Failure (parachute): 2; Precluded: 1

Data-quality note: launch-site naming is inconsistent; standardize labels before comparing counts.

SQL: examples of reproducible analysis

Queries turn raw records into interpretable summaries

First successful ground-pad landing

```
SELECT MIN(Date), Landing_Outcome
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

Payload mass total

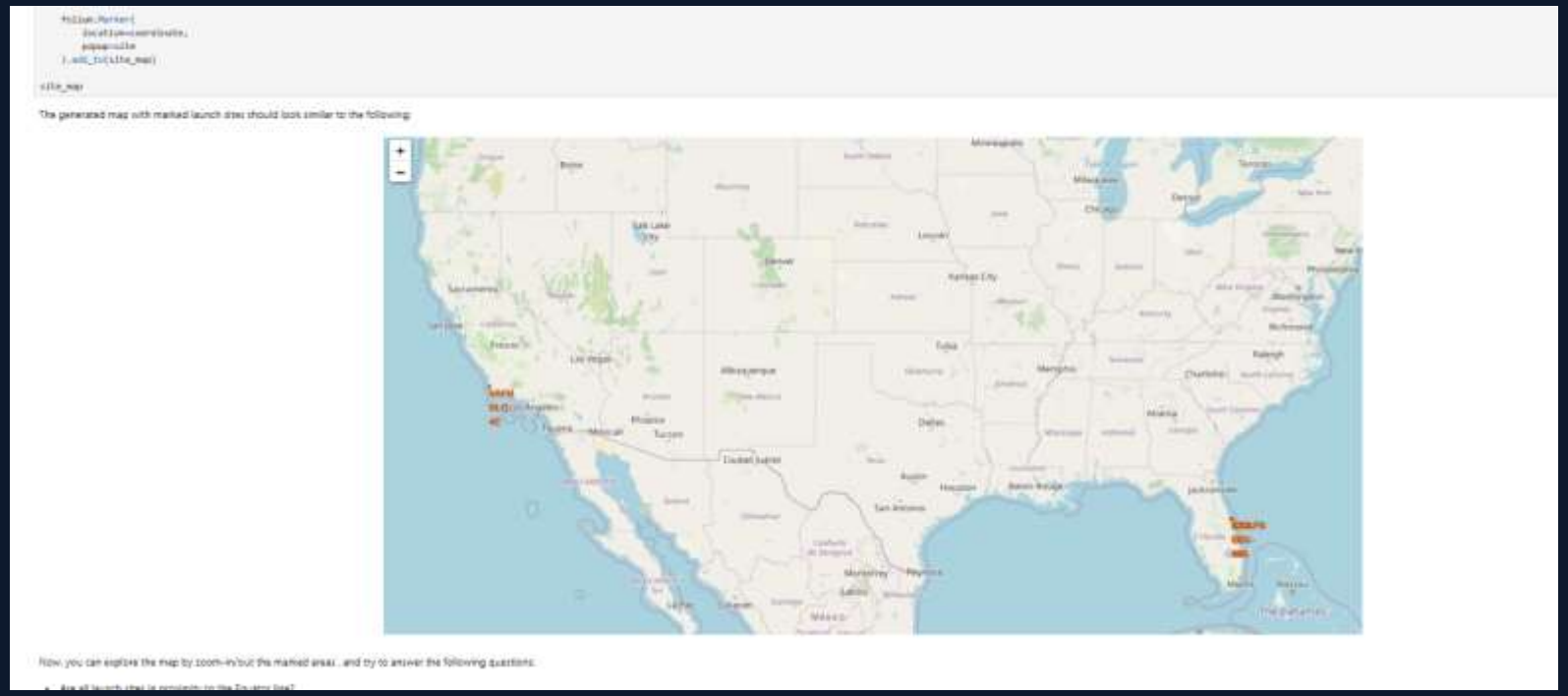
```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL;
```



Interpretation: SQL provides traceable answers, but query definitions and filters must be checked before drawing broader conclusions.

Geospatial analysis with Folium

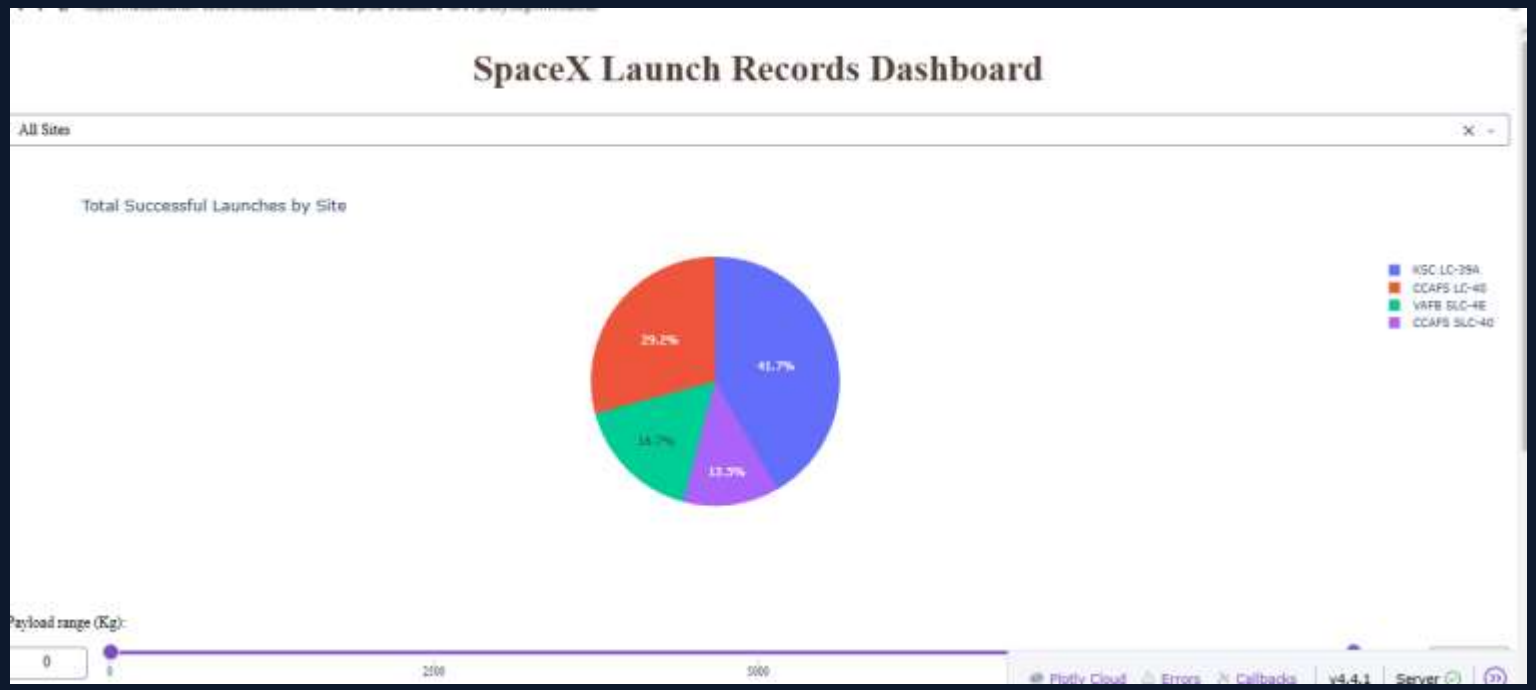
Launch sites are concentrated on the U.S. West and Gulf/East coasts



- Map markers identify launch-site locations in California and Florida.
- Geography provides context for comparing coastal launch facilities.
- Marker clustering improves readability when multiple records overlap.
- The map is descriptive; it does not by itself establish a relationship between location and landing success.

Interactive analysis with Plotly Dash

A working dashboard connects filters to two linked views





Dashboard interactions

- Filter by launch site.
- Adjust payload-mass range.
- Compare outcome and booster version.

Predictive analysis: classification

Cached notebook outputs — results should be revalidated on the corrected notebook

Model	Best CV accuracy	Test accuracy	Test correct
Logistic Regression	84.6%	83.3%	15 / 18
SVM	84.8%	83.3%	15 / 18
Decision Tree	89.1%	94.4%	17 / 18

- Decision Tree has the highest reported test accuracy in these cached outputs.
- The test set contains only 18 observations; one changed prediction shifts accuracy by 5.6 percentage points.
- Re-run the pipeline after fixing target separation, y shape, and confusion-matrix variable names before final submission.

Evaluation recommendation: include confusion matrices, precision/recall/F1, and a cross-validated comparison after rerunning the corrected pipeline.

Limitations & validation checks

What must be stated honestly in the final report

- Small sample: the classification notebook uses a test set of 18 observations, so metrics have high uncertainty.
- Potential leakage: scaling must be fitted only on training data; target Class must never be included in X.
- Reproducibility: the model scores above are cached outputs from the original notebook, not a verified end-to-end run of the corrected consolidated notebook.
- Data quality: launch-site spelling variants and mission outcome labels should be normalized before grouping.
- Dashboard/map screenshots demonstrate rendering, but the final repository should include runnable source code, dependency instructions, and a clear data-source description.

Conclusion & next steps

A practical, evidence-based summary

- Launch success patterns vary with orbit, flight number, payload mass, and launch site; no single visual feature cleanly separates outcomes.
- SQL, Folium, and Dash add complementary views: aggregate summaries, geographic context, and interactive filtering.
- The cached Decision Tree result is promising, but its small test set means it should not be presented as definitive.
- Next: rerun the corrected ML notebook, record final metrics and confusion matrices, normalize categories, and commit all deliverables to GitHub.

Core takeaway

The project demonstrates a complete analytical workflow—but model claims must remain proportional to the available evidence.

Repository & deliverables

Make the submission easy to review and reproduce

GitHub repository

https://github.com/MatiasMontenegrom/spacex-falcon9-landing-prediction

- Include the completed capstone notebook(s), the Dash application source, and any helper Python files.
- Add a README with setup instructions, data source, project sections, key findings, and how to run the dashboard.
- Include requirements.txt and avoid committing private credentials or large temporary files.
- Ensure the repository URL is clickable in the submitted PDF and that the repository is public or accessible to the grader.

Screenshots in this deck were supplied by the project author; model scores are explicitly marked as cached and pending revalidation.